

Explainable AI

Gentle Introduction to XAI methods.

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(Slides courtesy of Mahsa Dibaji and Susanne Schmid)

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Outline

- Introduction – What is explainability about?
- Taxonomy – Different XAI techniques
 - SHAP
 - GradCAM
 - Intrinsic Gradients
- Evaluation of XAI

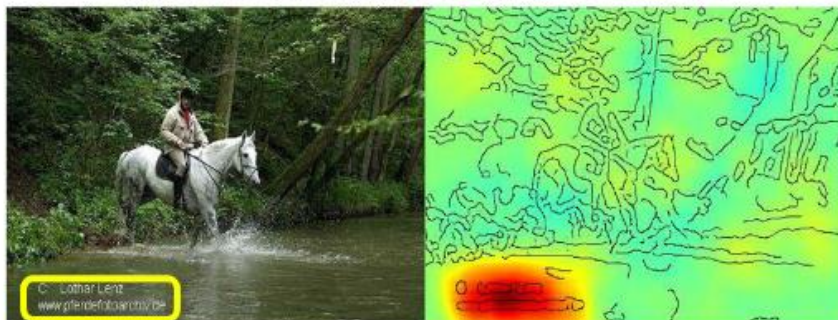
Learning Goals

- Get familiar with the purpose of XAI and the different categories model fall into.
- Get familiar with basics of some interpretability techniques.

Introduction

“Garbage in, Garbage out”

Horse-picture from Pascal VOC data set



Source tag
present



Classified
as horse

Artificial picture of a car



No source
tag present



Not classified
as horse



Introduction

Problem with “black box” model

Breast Cancer Detection Model Feng, 2020	
Accuracy rate:	0.85
Specificity:	0.857
Sensitivity:	0.847

Introduction

Importance of Explainability

1. Trust and Reliability

- Building Trust between user and AI system, Enabling User Feedback
- Risk management

2. Compliance and Ethical Considerations

- Making decision-process of AI systems more transparent

3. Model Debugging and Improvement

- Debug AI models
- Identify and mitigate biases

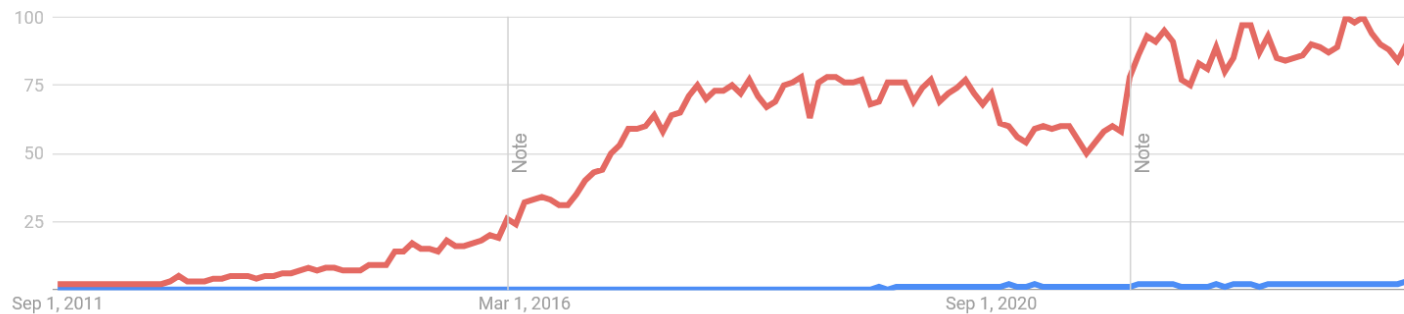
Introduction

- Explainability
 - Understanding the cause-and-effect relationships between inputs and outputs of a model. Focuses on the intuition behind model outputs.
- Interpretability
 - Understanding the internal logic and mechanics of a model. Focuses on the inner workings and processes of the model.

Explainable AI (Google trends)



Rising Interest in “Explainable AI”: A Google Trends Analysis Over Time (2011-2024)

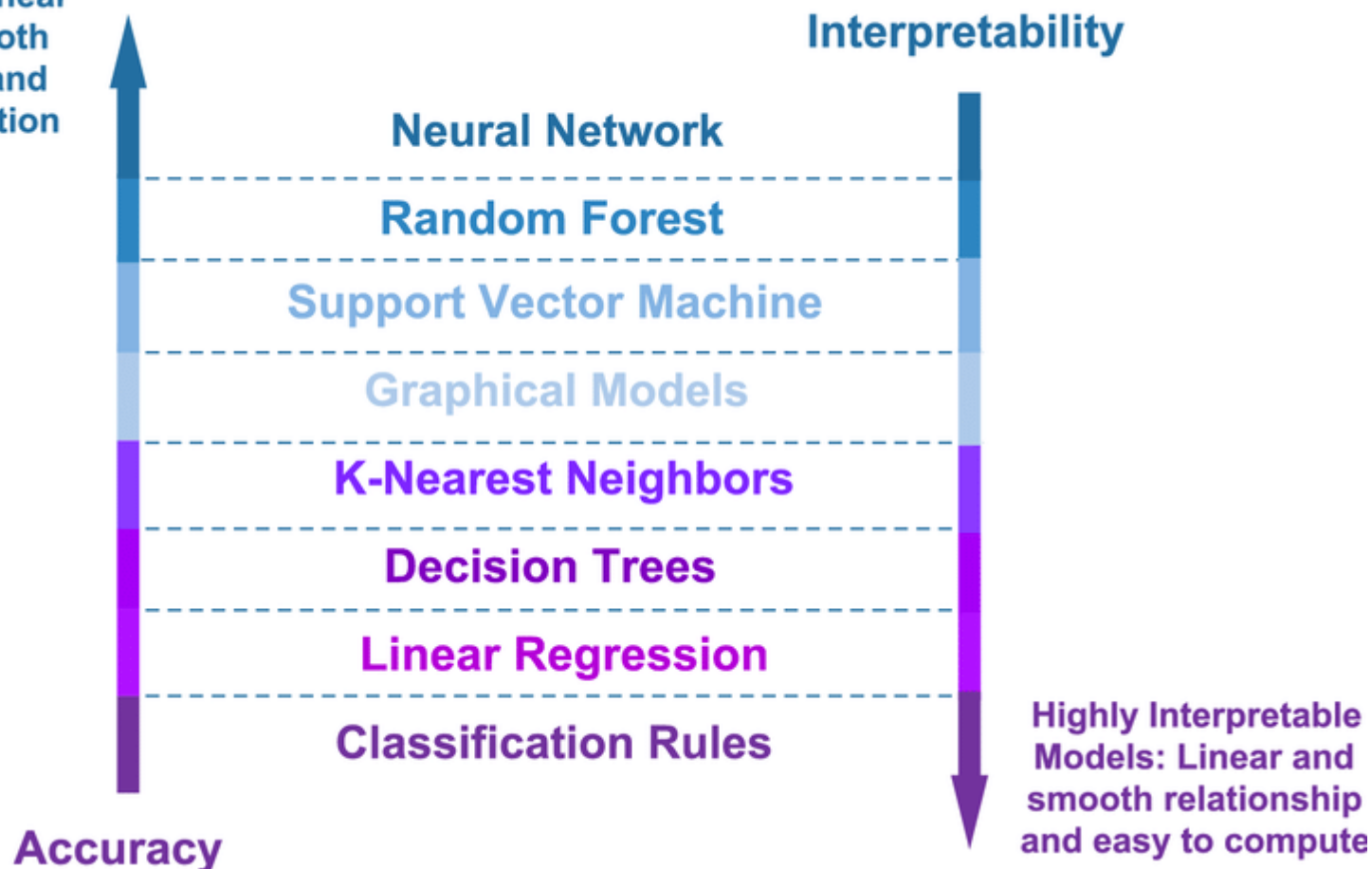


Deep Learning
Explainable AI

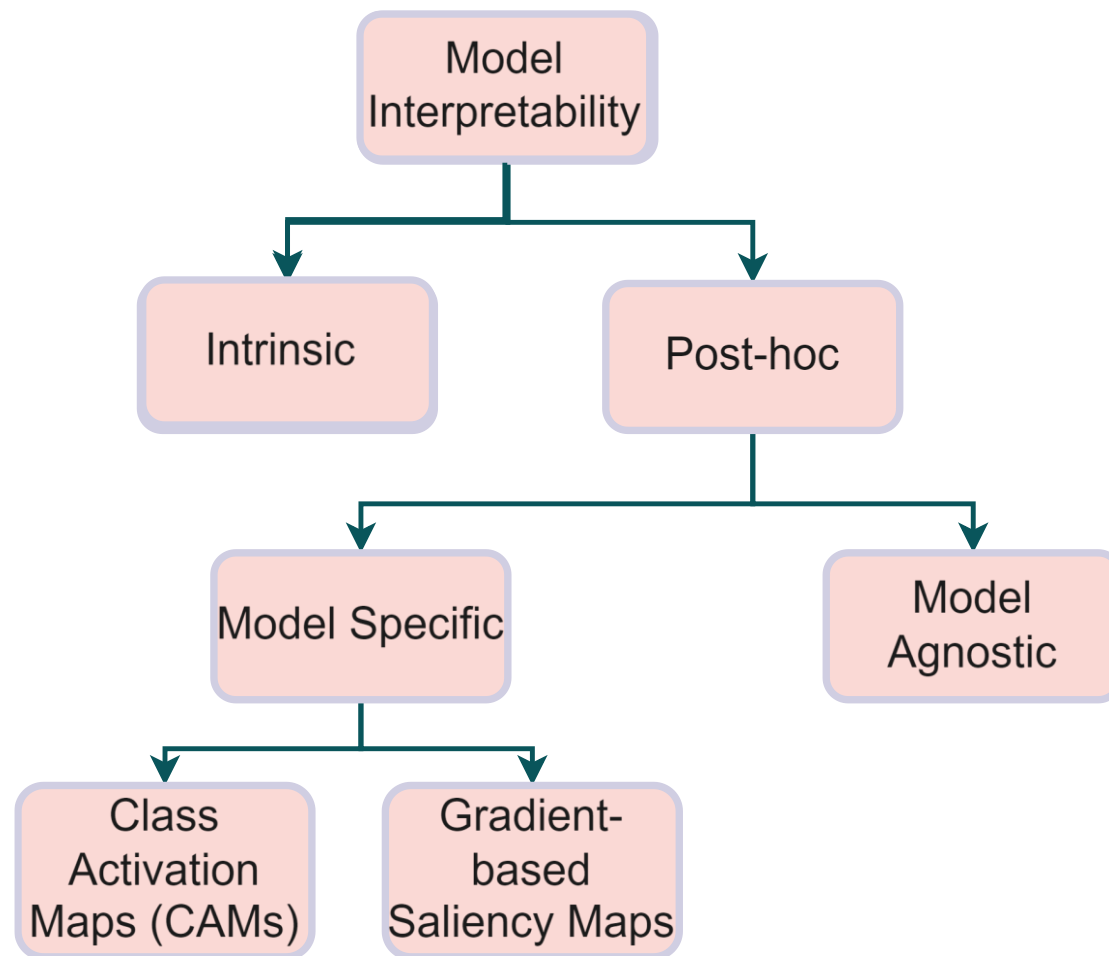
But still low compared to “deep learning”: A Google Trends Analysis Over Time (2011-2024)

Introduction

Highly Accurate
Models: Non linear
and Non smooth
relationship and
long computation
time

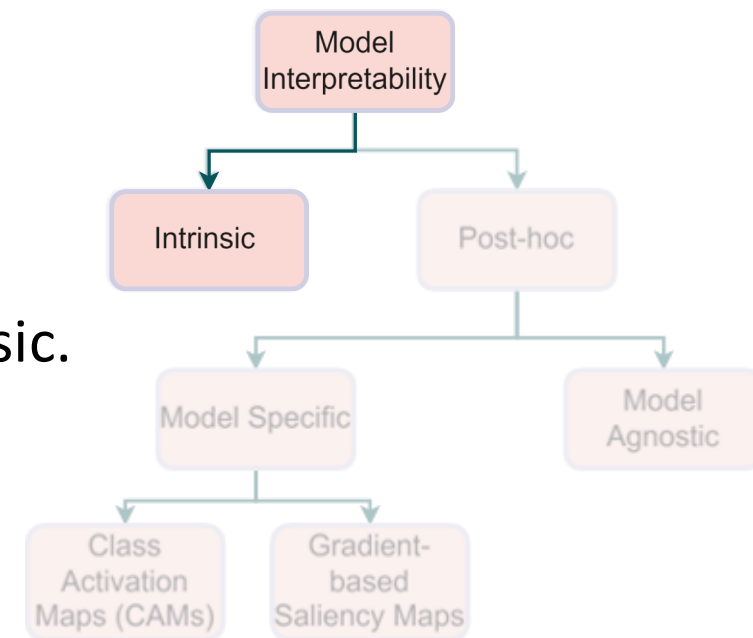


Interpretability Taxonomy



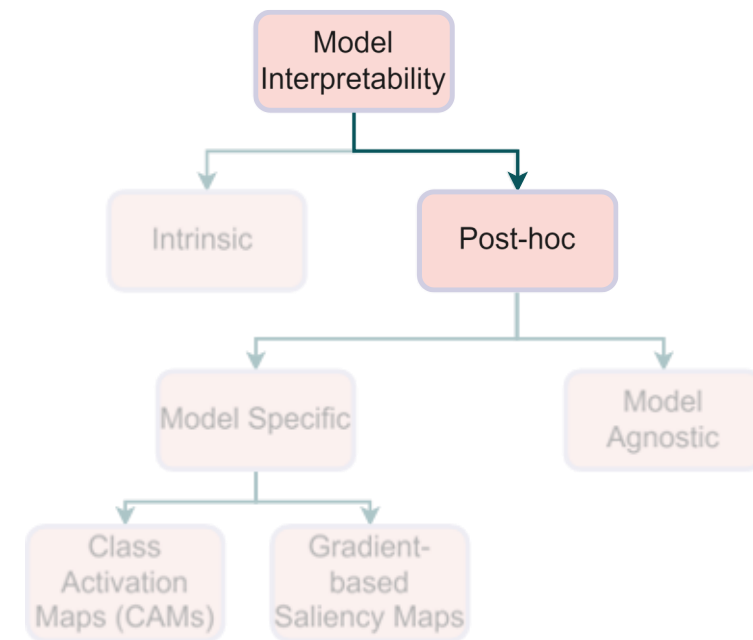
Intrinsic Interpretability

- The interpretability of ‘white-box’ models is defined as intrinsic.
- These models are interpretable by design.
- Examples
 - Rule-based systems such as decision trees.
 - A shallow neural network
 - Logistic regression
- Good choice for high-stakes decisions!



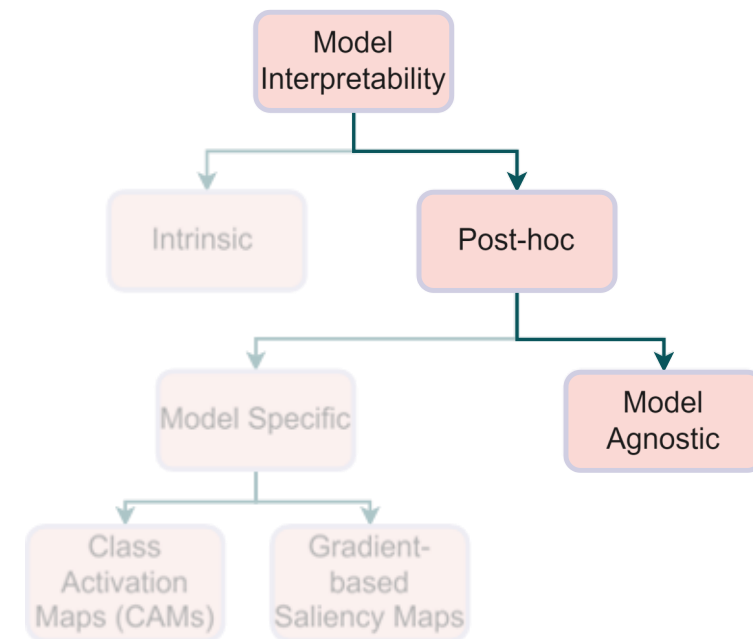
Post-hoc Interpretability

- Explaining a prediction of the model without explaining the exact internal mechanism of that model.
- Allows for the interpretation of complex models such as CNNs that are not inherently interpretable.
- Model-agnostic vs Model specific



Model-agnostic Techniques

- Techniques that can be applied to any model, regardless of its internal structure.
- To provide a way to interpret the decisions of complex models without needing access to their internal workings.
- Key Characteristics:
 - Independence from model type.
 - Focus on input-output relationships.
- Challenges: May not capture the intricacies of specific models, leading to less precise explanations.
- Examples:
 - LIME
 - SHAP
 - Perturbation-based methods



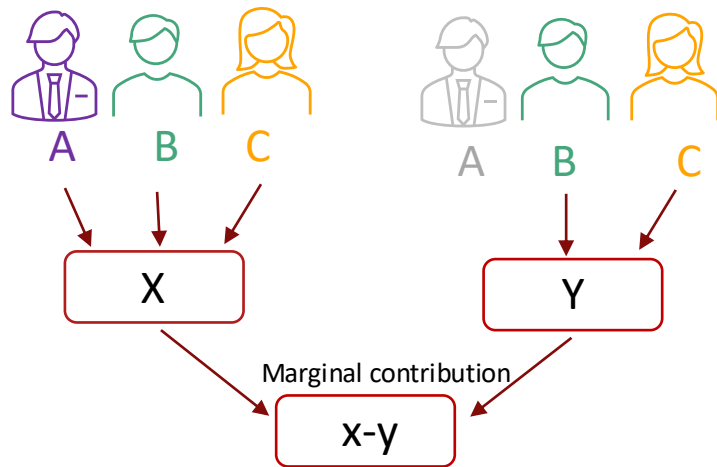
SHAP - SHapley Additive exPlanations

- **Shapley Values:** concept from cooperative game theory used to fairly distribute the total payoff (predictions) among players (features) based on their individual contributions.
- It quantifies the importance of each feature in model's decision-making process.

$$ShapV_i = \frac{1}{\# \text{ Coalitions}} \sum_{\text{All Coalitions}} \underbrace{(\text{Prediction with } F_i - \text{Prediction without } F_i)}_{\text{Marginal contributions}}$$

Coalitions: A subset of features working together to influence the prediction.

SHAP - SHapley Additive exPlanations

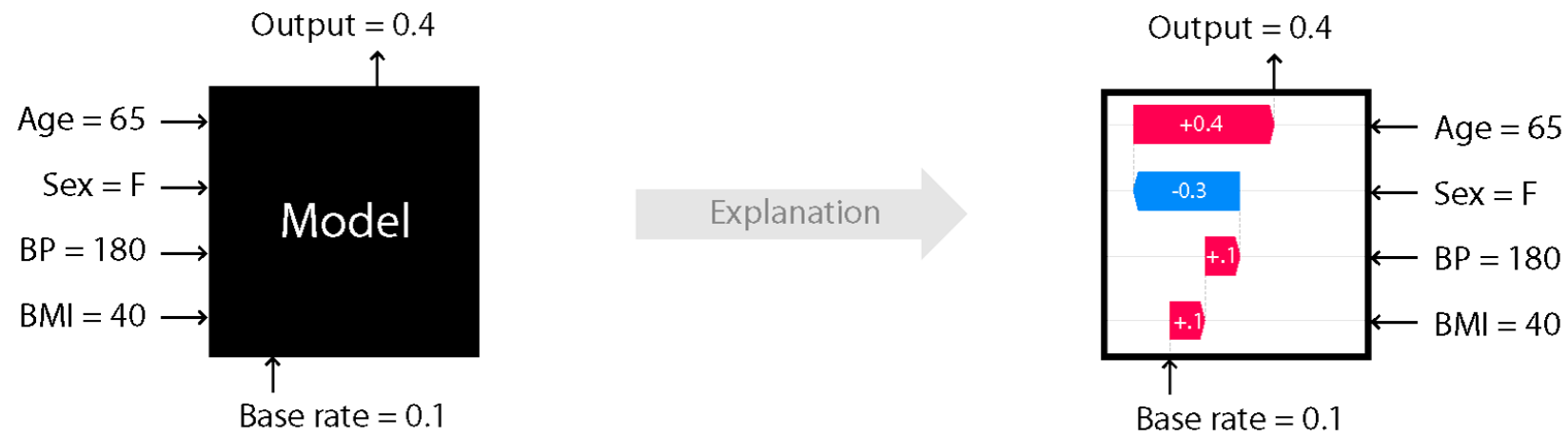


	Cost
A	\$50
B	\$60
C	\$80
AB	\$70
BC	\$90
AC	\$100
ABC	\$120

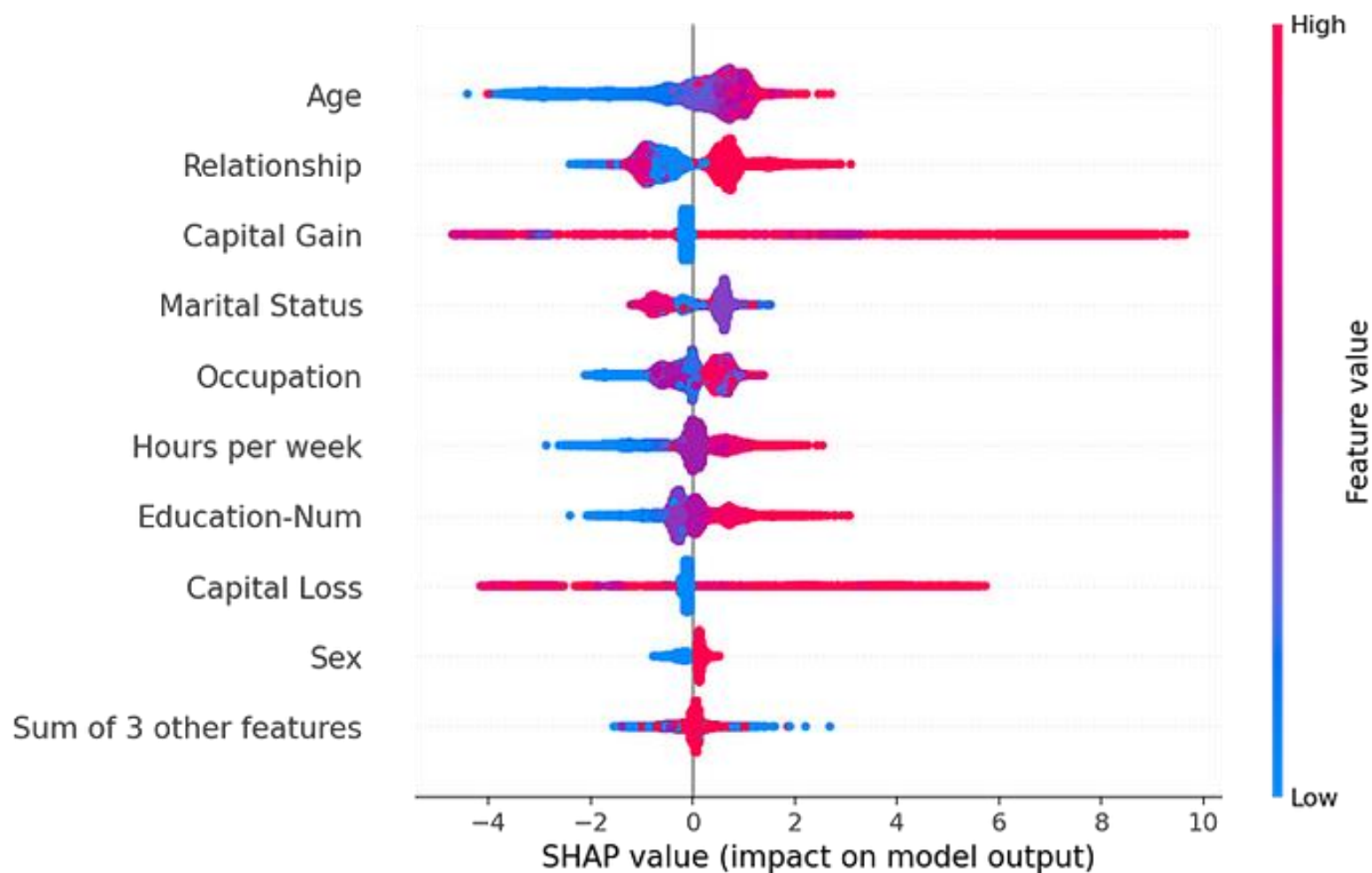
Constellation	A	B	C
ABC	\$50	\$20	\$50
BAC	\$10	\$60	\$50
ACB	\$50	\$20	\$50
BCA	\$30	\$60	\$30
CAB	\$20	\$20	\$80
CBA	\$30	\$10	\$80
Average	31.7	31.7	56.7

1. **Efficiency:** the final reward must be shared among all
2. **Symmetry:** players who made the same contribution will receive the same amount of reward.
3. **Dummy:** players who did not contribute will receive no reward.
4. **Additivity:** Shapley values for different predictions can be added together

SHAP - SHapley Additive exPlanations



Example on Tabular Data (Adult Census Income)



- Magnitude:** A larger absolute SHAP value a larger impact on the model's output.

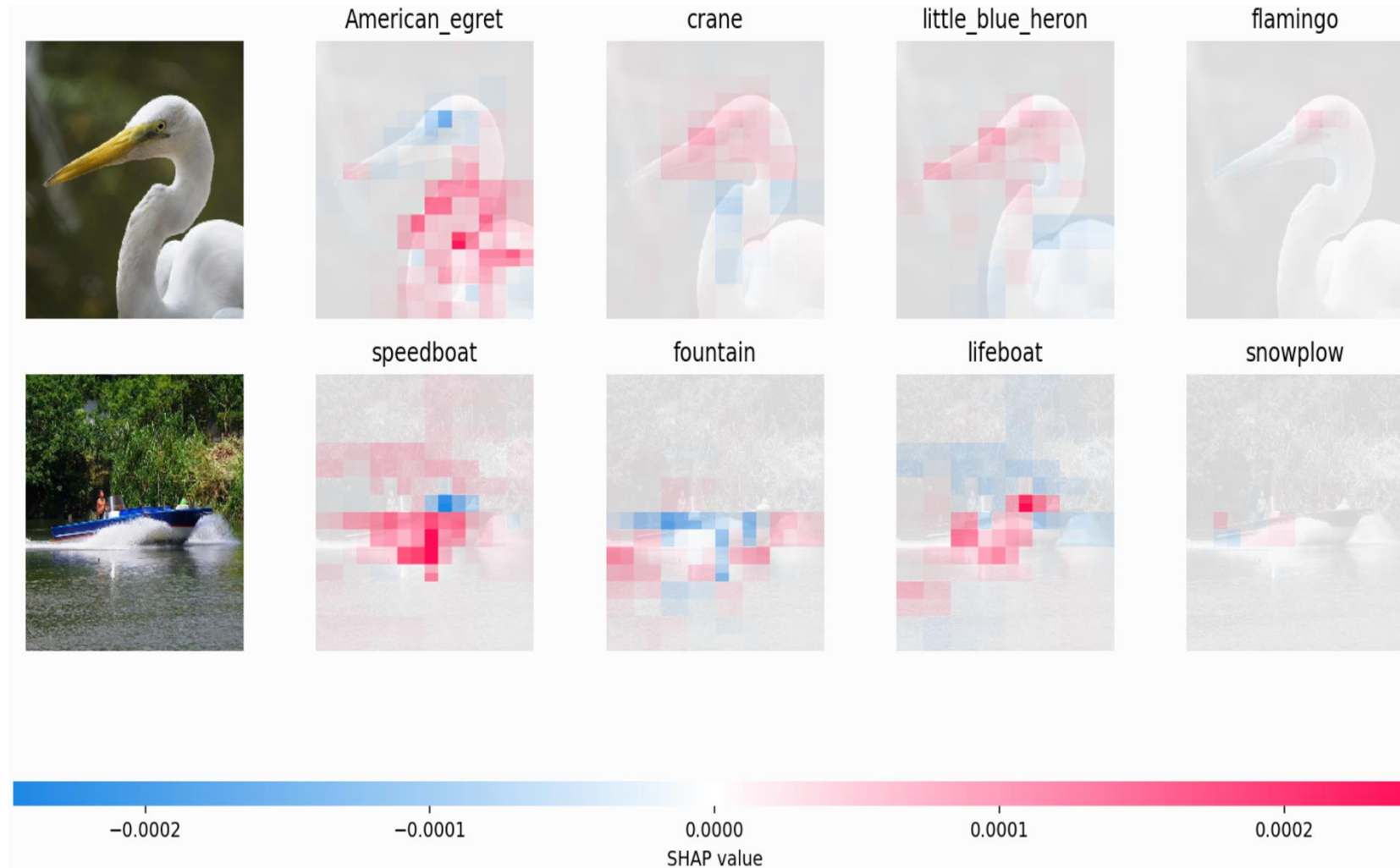
- Sign:** whether the effect of that feature is to increase or decrease the prediction.

- Color:** Value of the feature

How SHAP works?

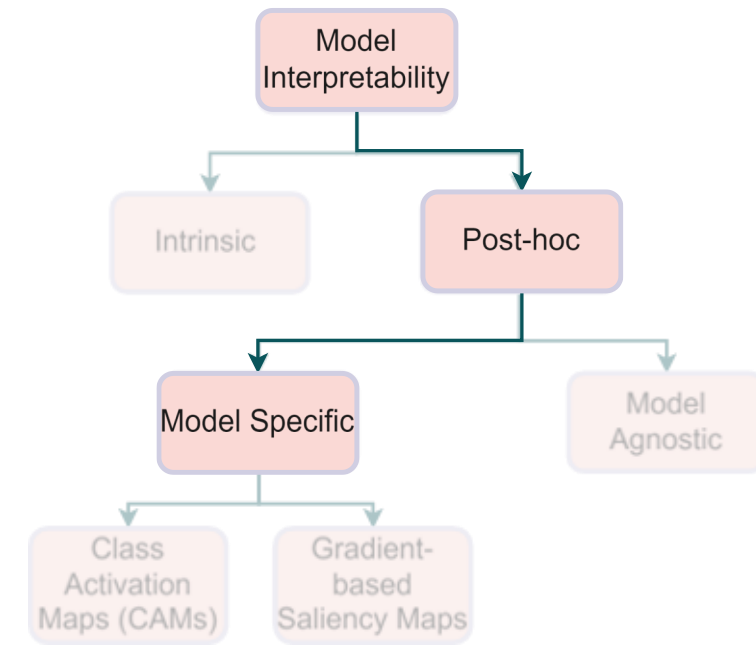
1. **Select a prediction to explain:** Choose an instance from your dataset that has been run through the model to get a prediction.
 2. **Create a baseline output:** The baseline is typically the average prediction over the dataset, representing the prediction that would be made without any information about the current instance.
 3. **Compute SHAP Values:**
 - For each feature, consider all possible sets of features (coalitions) that could include that feature. Compute the change in prediction with and without the feature across these sets.
 - The average of these differences across all coalitions is the SHAP value for that feature.
 - The SHAP value represents the feature's average marginal contribution to the prediction compared to the baseline.
- Computation of Shapley values is complex for models with large number of features.
- Different SHAP explainers optimize computation of shapley values for different types of ML/DL.

Example on Image Data (ImageNet) – Deep SHAP



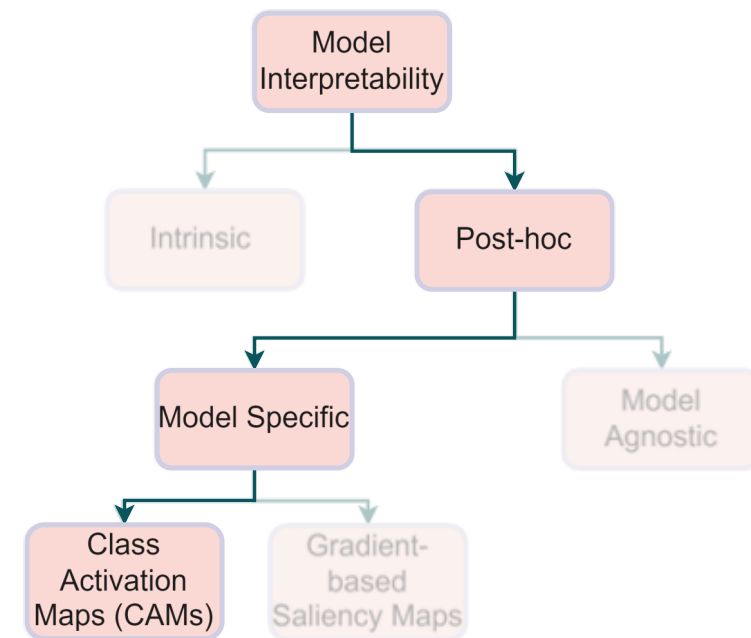
Model-specific Techniques

- Techniques designed to explain specific types/groups of models, utilizing their internal structure.
- leverage internal model structure of certain models to provide more detailed and accurate explanations.
- Key Characteristics:
 - Tailored to specific model architectures.
 - Can provide insights into the internal decision-making process of the model.
- Challenge: Limited in applicability to other model types.
- Examples:
 - Gradient-based saliency maps: Vanilla Gradient, Integrated Gradients, SmoothGrad
 - Class Activation Maps: GradCAM, GradCAM++
 - LRP

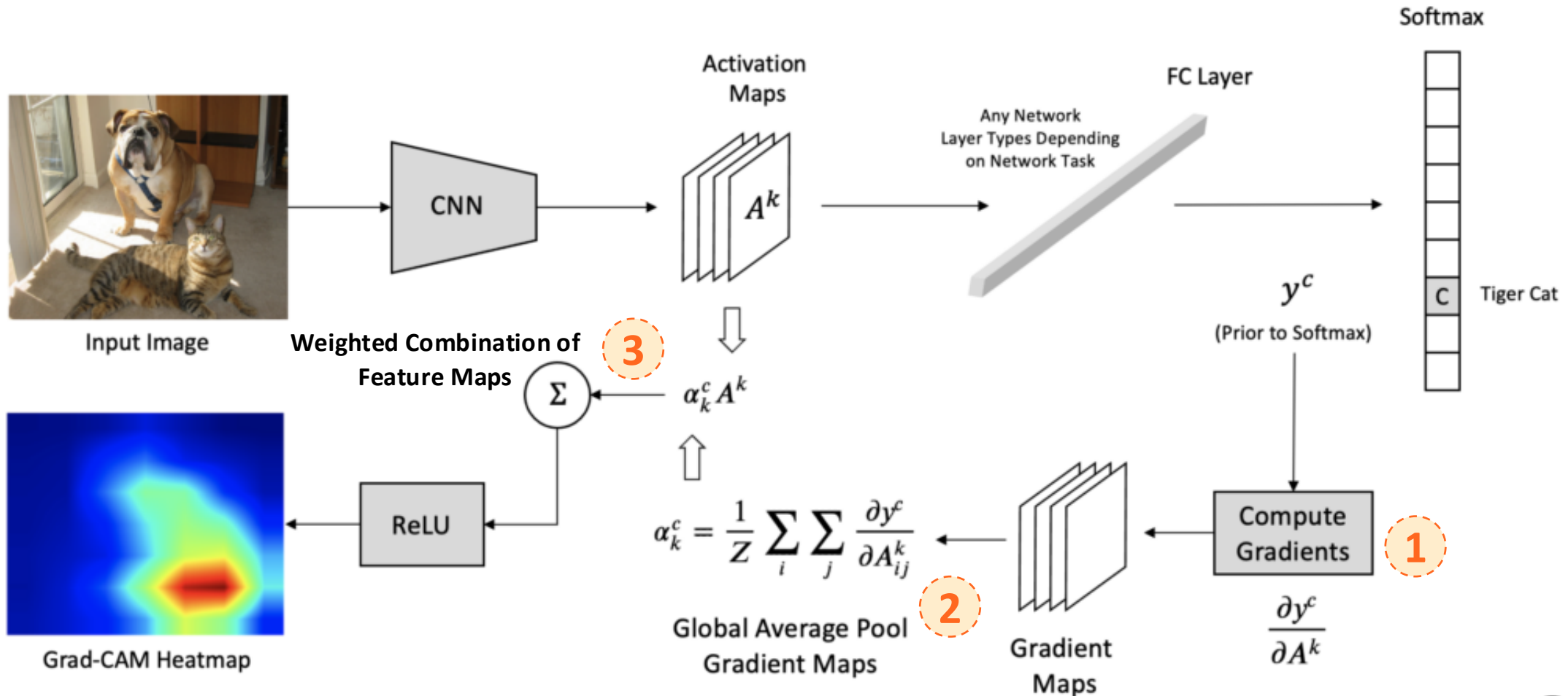


GradCAM

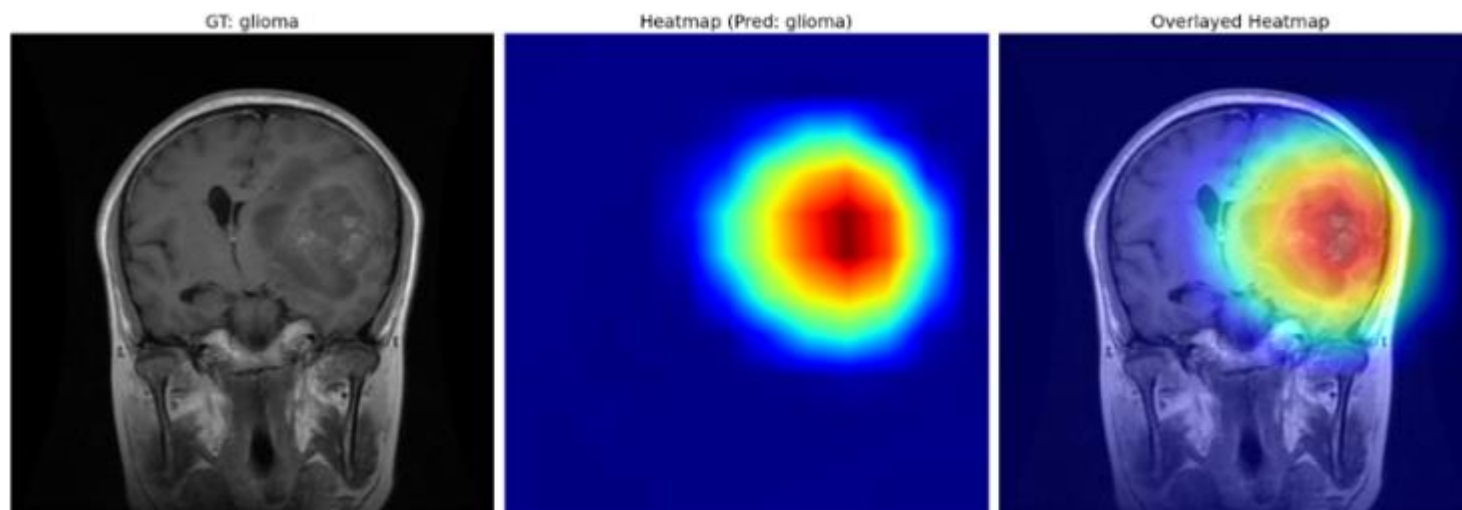
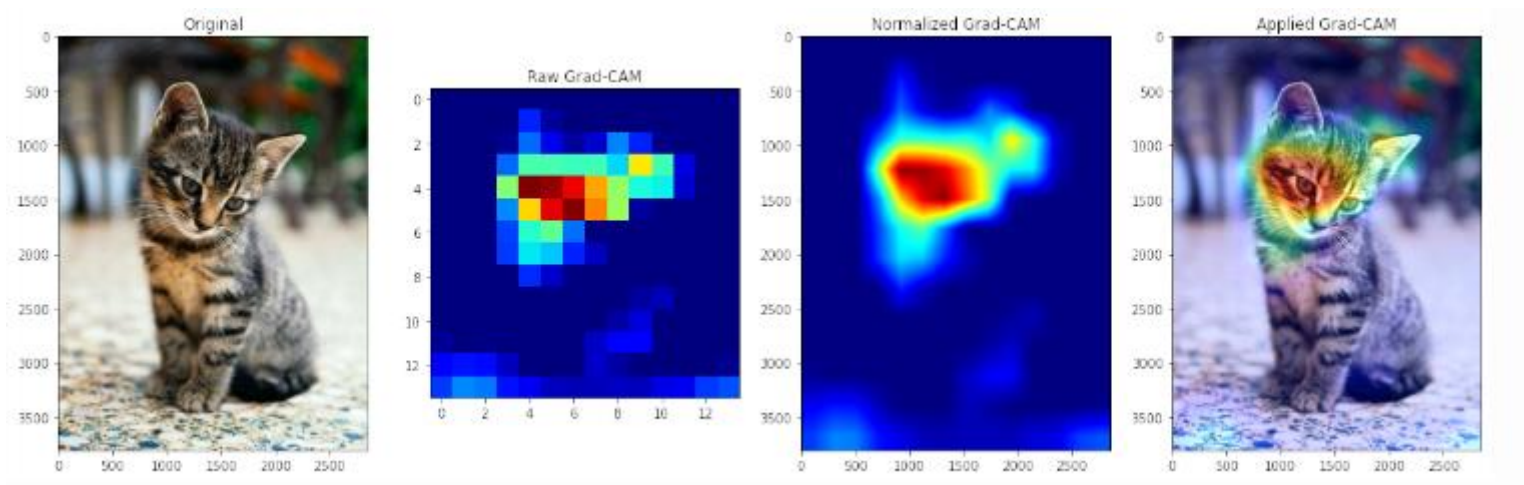
- Gradient-weighted Class Activation Mapping.
- A Generalization over CAM that can be applied to a wider range of convolutional neural network (CNN)
- Produce a coarse localization map highlighting the important regions in the image for making a prediction.



How GradCAM works?

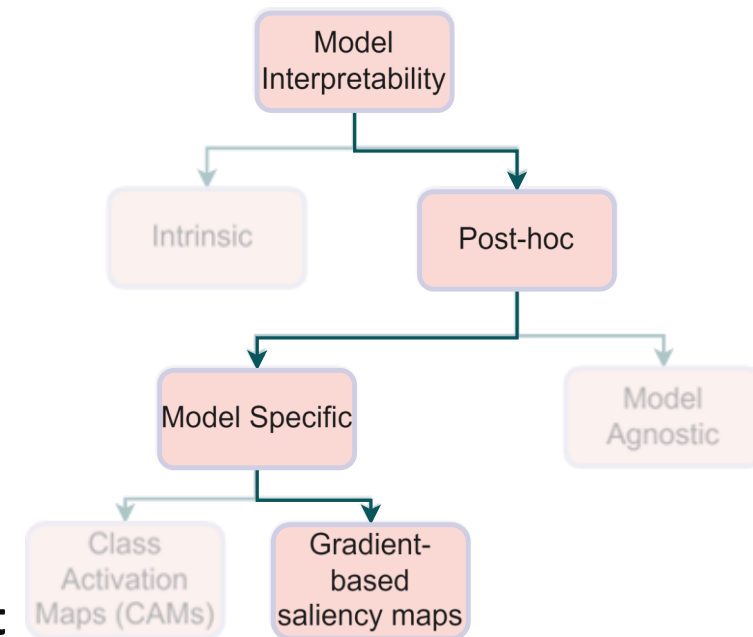


GradCAM - Example



Integrated Gradients

- Integrated gradients aims to attribute the prediction of **any differentiable neural network** to its input features.
- Considers the **path** from a **baseline input** to the **actual input** and **computing the gradients** of the model's **output with respect to the input** along this path.
- Gradients provide information about how much each feature contributes to the change in the model's prediction as you move from the baseline to the actual input.

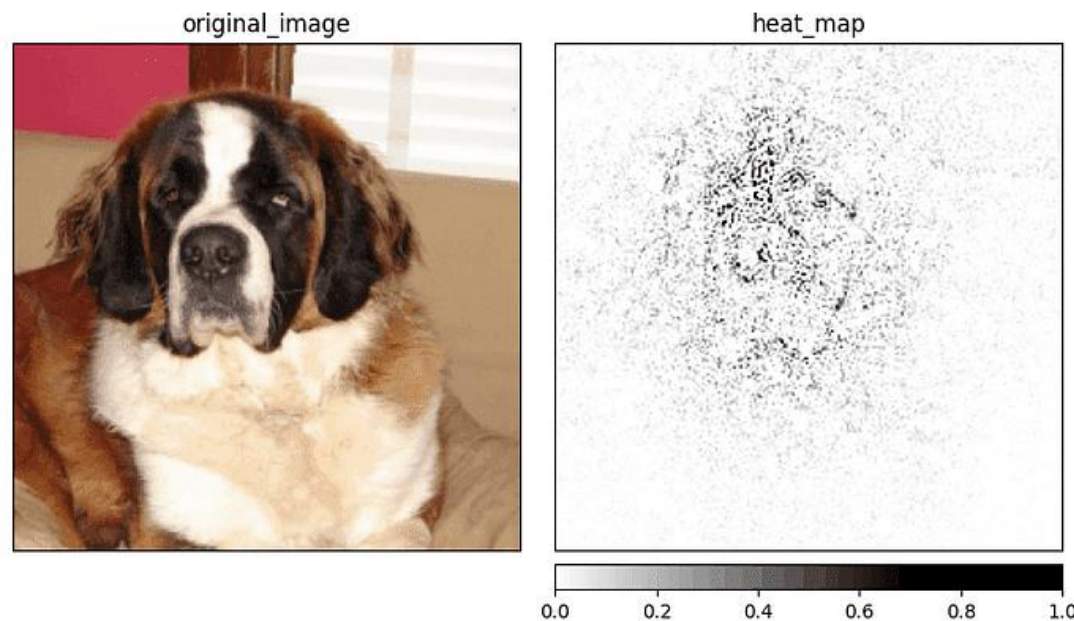


Integrated Gradients: Example



baseline

Actual Input



How Integrated Gradients works?

1. **Baseline Input:** An input that represents the absence of features or a neutral starting point; e.g., a black image.
2. **Path from Baseline to Actual Input:** Usually linearly interpolating between the baseline and the actual input.
3. **Compute Gradients Along the Path:** Gradients of output w.r.t each input.
4. **Integrate the Gradients:** Aggregates the contribution of each feature across the path.
5. **Attribution:** Integrated gradients along i^{th} dimension represent i^{th} feature's attribution (contribution to prediction)

$$\text{IntegratedGrads}_i^{\text{approx}}(x) ::= (x_i - x'_i) \times \sum_{k=1}^m \frac{\partial F(x' + \frac{k}{m} \times (x - x'))}{\partial x_i} \times \frac{1}{m}$$

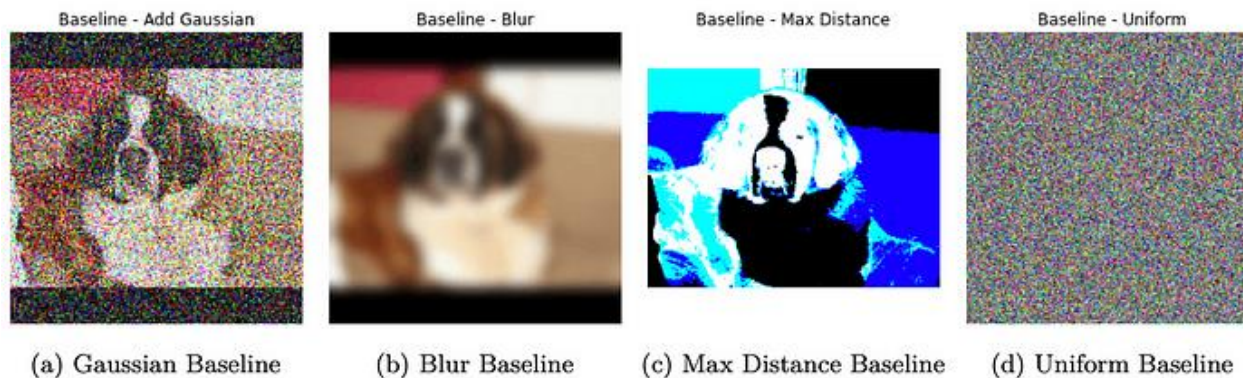
x : Actual Input x' : baseline input

m : number of interpolations

$F : \mathbb{R}^n \rightarrow [0, 1]$ is the deep network

Integrated Gradients - Additional notes

- Based on two axioms:
 - **Sensitivity:** *An attribution method satisfies **Sensitivity** if for every input and baseline that differ in one feature but have different predictions, then the differing feature should be given a non-zero attribution. If the function implemented by the deep network does not depend (mathematically) on some variable, then the attribution to that variable is always zero.*
 - **Implementation Invariance:** *Two networks are **functionally equivalent** if their outputs are equal for all inputs, despite having very different implementations. Attribution methods should satisfy **Implementation Invariance**, i.e., the attributions are always identical for two functionally equivalent networks.*
- Choice of Baseline: May affect the attributions and is still an open-question (potential research focus! 😊)



Summary

- Explainable AI techniques helps us understand the decision-making process of ML/DL models.
- Interpretability is either intrinsic, or we can achieve it with post-hoc methods.
- Post-hoc interpretability methods are model specific or model agnostic.

References

- <https://medium.com/@kemalpiro/xai-methods-integrated-gradients-6ee1fe4120d8>
- <https://towardsdatascience.com/xai-methods-the-introduction-5b1b81427c9c>
- <https://www.mdpi.com/1099-4300/23/1/18>
- <https://medium.com/@shahooda637/all-you-need-to-know-about-shap-for-explainable-ai-8ad35a05e6ec>
- <https://learnopencv.com/intro-to-gradcam/>

More resources

- <https://christophm.github.io/interpretable-ml-book/>
- <https://github.com/jphall663/awesome-machine-learning-interpretability>
- <https://github.com/Trusted-AI/AIX360>
- <https://github.com/interpretml/interpret>